

AI-Driven IoT Smart Fisherman Safety System for Real-Time Emergency Detection

SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing

Project ID
J26_IT_302

Project Proposal Report
Kumarathunga R.C.S – IT23257054

**B.Sc. (Hons) Degree in Information Technology Specialized in Information
Technology**

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

September 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Kumarathunga R.C.S	IT23257054	

The above candidate is conducting research for their undergraduate Dissertation under my supervision

Name	Date	Signature
Prof. Samantha Rajapaksa	2026/08/14	

ABSTRACT

Fishing is an important economic activity in Sri Lanka and provides income for many coastal communities. However, small-scale fishermen operate in hazardous and unpredictable marine environments where incidents such as sudden weather changes, boat capsizing, water leakage, engine failures, onboard fires, and accidental falls overboard can occur. Existing emergency communication approaches commonly depend on mobile phones or conventional radio systems, which require manual operation and may become unreliable in offshore areas with limited network connectivity. Furthermore, existing IoT-based fisherman safety solutions primarily emphasize GPS tracking and emergency communication, with limited support for continuous onboard condition monitoring and intelligent local processing.

This research proposes the **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing** as the embedded hardware component of the overall SeaGuardian system. The proposed device uses an ESP32 microcontroller integrated with a NEO-6M GPS module for location tracking, MPU6050 for boat tilt and motion monitoring, water leakage detection, DS18B20 temperature sensing, battery monitoring, and a manual SOS mechanism with buzzer and LED alerts. MicroSD-based offline storage and edge data processing will allow sensor information to be processed and retained locally when internet connectivity is unavailable.

The component will be developed and evaluated through controlled laboratory simulations, sensor calibration, hardware testing, and simulated emergency conditions. Evaluation will consider sensor functionality and accuracy, response behaviour, power consumption, and overall system reliability. The resulting IoT device will provide real-time onboard sensor information to the wider SeaGuardian platform, supporting communication, cloud synchronization, Digital Twin visualization, AI-based risk assessment, mobile monitoring, and emergency response functions.

Keywords: Internet of Things (IoT), Fisherman Safety, Edge Processing, ESP32, Emergency Detection

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my research supervisor for the continuous guidance, valuable feedback, encouragement, and support provided throughout the planning and development of this research project.

I am also grateful to the academic staff of the Sri Lanka Institute of Information Technology (SLIIT) for providing the knowledge, resources, and academic environment necessary to conduct this research.

I would like to thank my fellow research group members for their cooperation, discussions, technical support, and contribution towards the development of the complete SeaGuardian system. Their collaboration has been valuable in defining the integration between the IoT device, communication infrastructure, intelligent risk analysis, mobile rescue platform, fish storage monitoring, and boat health monitoring components.

Finally, I would like to express my appreciation to my family and everyone who provided encouragement and support throughout this research journey.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
AIMS	Autonomous Intelligent Machines and Systems
API	Application Programming Interface
ESP32	Espressif 32-bit Microcontroller
GPS	Global Positioning System
IoT	Internet of Things
LED	Light Emitting Diode
MPU	Motion Processing Unit
RF	Radio Frequency
SDG	Sustainable Development Goal
SOS	Emergency Distress Signal
WBS	Work Breakdown Structure

LIST OF APPENDICES

Appendix	Title
Appendix A	Proposal Evidence and Decision Log
Appendix B	Hardware Purchase Receipts and Cost Evidence
Appendix C	System Architecture and Hardware Design
Appendix D	Prototype Development and Hardware Integration Evidence
Appendix E	Experimental Test Plan
Appendix F	Stakeholder / User Requirement Evidence
Appendix G	Literature Comparison
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1. INTRODUCTION

1.1 Background and Context

Fishing is an important economic activity that supports the livelihoods of many coastal communities. However, fishing operations are associated with significant occupational and maritime safety risks because fishermen work in dynamic and unpredictable marine environments. Previous research has highlighted the importance of improving safety, navigation, communication, and information-sharing technologies for fishing vessels [1].

Small-scale fishing boats commonly rely on conventional communication and location-based mechanisms for safety and emergency support. Existing IoT-based solutions have investigated technologies such as GPS, GSM, LoRa, and other wireless communication mechanisms to provide vessel tracking, maritime-border alerts, and emergency information [2]–[5]. However, such approaches commonly emphasize location tracking, border monitoring, and communication functionality rather than comprehensive onboard condition monitoring.

Existing research has explored the use of Internet of Things (IoT), Global Positioning System (GPS), and wireless communication technologies for fisherman safety. Such systems demonstrate the potential of IoT technologies to improve location tracking, emergency communication, and the transmission of safety information. Nevertheless, the existing approaches identified during the initial research primarily concentrate on GPS tracking, maritime border monitoring, and emergency communication. Comprehensive onboard monitoring involving boat motion, water leakage, temperature, battery condition, and emergency controls together with local edge processing and offline data storage remains insufficiently addressed.

To address this problem, the overall **SeaGuardian** solution is proposed as an AI-driven IoT-based smart fisherman safety and emergency communication system. The complete system combines an onboard IoT device, hybrid emergency communication, cloud services, Digital Twin visualization, AI-based risk assessment, mobile rescue functionality, boat health monitoring, and fish storage monitoring. These components are intended to work together to support real-time monitoring, intelligent risk assessment, and emergency response.

This individual research focuses specifically on the **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing**. The proposed component uses an ESP32 as the main embedded controller and integrates a NEO-6M GPS module for location tracking, an MPU6050 for boat tilt and motion monitoring, water leakage detection, DS18B20 temperature sensing, battery monitoring, an SOS button, and buzzer and LED alert mechanisms. MicroSD-based offline storage and local edge processing are also proposed so that the device can continue collecting, processing, and retaining important sensor information when internet connectivity is unavailable.

The proposed individual component forms the primary onboard data-acquisition and edge-processing layer of SeaGuardian. Sensor information generated by the device can subsequently be transferred through the communication infrastructure to the other components of the complete solution for cloud synchronization, Digital Twin visualization, AI-based risk analysis, mobile monitoring, and emergency response. Therefore, the reliability of the onboard IoT device is important to the operation of the overall SeaGuardian platform.

The proposed research is aligned with the **Autonomous Intelligent Machines and Systems (AIMS)** research cluster. It also contributes to relevant United Nations Sustainable Development Goals, particularly **SDG 3 – Good Health and Well-being**, through improving safety and emergency response; **SDG 9 – Industry, Innovation and Infrastructure**, through the application of IoT and intelligent technologies; and **SDG 14 – Life Below Water**, through supporting safer fishing activities and maritime operations.

1.2 Literature Review

Existing research demonstrates that Internet of Things (IoT), Global Positioning System (GPS), and wireless communication technologies can support the safety and monitoring of fishermen operating in marine environments. Previous research has investigated vessel location tracking, maritime-border detection, emergency alert mechanisms, wireless communication, and intelligent monitoring approaches [1]–[7].

Ramesh *et al.* [3] investigated an IoT-assisted fisherman-support approach using intelligent GPS technology for border detection and alerting. The proposed approach demonstrates the usefulness of GPS and IoT technologies for identifying vessel location and supporting fishermen when approaching restricted maritime boundaries. While this type of solution improves location awareness and safety notification, its primary emphasis is GPS-based border monitoring rather than integrated monitoring of multiple onboard physical conditions.

Similarly, the fisherman border-detection approach presented in [2] uses IoT and LoRa technologies to support maritime-border detection and alerting. The study demonstrates the potential of long-range wireless communication and location information for fisherman-safety applications. However, its principal functionality is related to geographical border detection and communication rather than continuous monitoring of boat motion, water leakage, temperature, battery condition, and other onboard safety parameters.

Gomathi *et al.* [4] proposed an SMS- and GPS-based fisherman tracking system using IoT. The approach demonstrates how GPS information and communication technologies can be combined to support fisherman tracking and safety. Jothilingam and Glindis [5] similarly investigated an IoT-based border-alert and secured system for fishermen. These approaches further demonstrate the strong emphasis placed on location, tracking, border awareness, and communication in existing fisherman-safety research.

A broader review by Amuthakkannan *et al.* [1] examined fisherman-boat automation involving artificial intelligence and technologies supporting safety, security, navigation, and information sharing. The review demonstrates the increasing potential for intelligent and connected technologies to improve fishing operations and fisherman safety. It also provides motivation for investigating more integrated onboard sensing and intelligent processing architectures.

More recent maritime-safety research has investigated intelligent multi-sensor monitoring. Yang *et al.* [6], for example, investigated deep-learning-based fisherman behaviour recognition using wearable sensing technologies. Related work has also investigated multimodal sensing and online learning for onboard crew-behaviour prediction [7]. Although these approaches demonstrate the potential of sensor fusion and intelligent processing in maritime environments, their primary focus is human behaviour monitoring rather than the integrated monitoring of fishing-boat safety conditions addressed by this research.

The reviewed literature therefore demonstrates significant progress in GPS-based tracking, border detection, wireless emergency communication, and intelligent maritime monitoring. Nevertheless, comparatively limited attention has been identified in the reviewed studies toward combining GPS location, boat tilt and motion, water leakage, temperature, battery condition, manual emergency activation, local alert mechanisms, edge processing, and offline data retention within a single low-cost embedded device intended for small-scale fishing environments.

Based on these observations, the proposed SeaGuardian IoT Device investigates an ESP32-based onboard platform integrating a NEO-6M GPS module, MPU6050 motion sensor, water leakage detection, DS18B20 temperature sensing, battery monitoring, SOS emergency control, buzzer and LED alerts, local edge processing, and MicroSD-based offline storage. The component is intended to operate as the onboard sensing and processing layer of the wider SeaGuardian system.

1.3 Research Gap

Existing IoT-based fisherman safety systems have frequently focused on GPS-based location tracking, maritime-border monitoring, and emergency communication [2]–[5]. These approaches demonstrate the usefulness of IoT and wireless technologies for transmitting location information and supporting safety alerts. However, the reviewed systems provide comparatively limited support for comprehensive and continuous monitoring of multiple onboard safety conditions within a single embedded device.

In particular, monitoring parameters such as boat tilt and motion, water leakage, temperature, battery condition, and manual emergency activation are not comprehensively integrated with local edge processing in the identified existing approaches. Many connected safety systems also depend on continuous communication with external platforms for data transmission and processing. Such dependency can become a limitation in offshore environments where internet or mobile network connectivity may be unreliable or unavailable.

Another identified limitation is the lack of offline data-handling capabilities within the onboard safety device. If connectivity is interrupted, sensor information generated during a potentially dangerous situation should still be collected and retained. Local processing is also important because immediate onboard safety functions should not depend entirely on communication with a remote server.

Therefore, the specific research gap addressed by this individual study is the **limited availability of a low-cost, integrated onboard IoT safety device that combines multi-sensor monitoring, local edge processing, emergency alert mechanisms, and offline data storage for small-scale fishing environments with intermittent connectivity**.

To address this gap, the proposed SeaGuardian IoT Device integrates an ESP32 with GPS location tracking, boat tilt and motion monitoring, water leakage detection, temperature sensing, battery monitoring, an SOS emergency mechanism, local buzzer and LED alerts, MicroSD-based offline storage, and edge data processing. The research will investigate the feasibility and reliability of integrating these capabilities into a single embedded platform that operates as the onboard sensing and data-processing layer of the complete SeaGuardian system.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a low-cost ESP32-based SeaGuardian IoT device for real-time onboard monitoring, local edge processing, emergency detection, and offline data storage to improve the safety monitoring capabilities of small-scale fishing boats operating under limited or intermittent network connectivity.

2.2 Specific Objectives

1. **To design and develop an ESP32-based embedded IoT device** capable of integrating multiple onboard safety sensors and emergency control mechanisms into a single platform.
2. **To implement real-time onboard monitoring** by integrating a NEO-6M GPS module for location tracking, an MPU6050 for boat tilt and motion monitoring, a water leakage sensor, a DS18B20 temperature sensor, and a battery monitoring mechanism.
3. **To develop local edge-processing and emergency-response functionality** capable of processing and filtering sensor readings locally and activating appropriate buzzer and LED alerts together with a high-priority manual SOS mechanism.
4. **To implement MicroSD-based offline data storage** to retain important sensor and emergency information when network connectivity is unavailable and support subsequent synchronization with the wider SeaGuardian system.
5. **To design and assemble the IoT device within a suitable waterproof enclosure** and perform sensor calibration and individual hardware testing before integration with the SeaGuardian communication infrastructure.
6. **To evaluate the developed IoT device** through controlled laboratory simulations by assessing sensor functionality and accuracy, emergency response behaviour, power consumption, and overall system reliability under simulated operating and emergency conditions.

3. METHODOLOGY

3.1 Research Methodology / Research Design

This research will adopt an **experimental research design supported by iterative prototype development** to design, implement, and evaluate the SeaGuardian IoT Device. This approach is appropriate because the research focuses on developing a physical embedded IoT prototype and experimentally evaluating its behaviour under controlled and simulated onboard conditions.

The research process will begin with the selection and individual testing of the required hardware components. An ESP32 microcontroller will operate as the main processing unit, while a NEO-6M GPS module, MPU6050 motion sensor, water leakage sensor, DS18B20 temperature sensor, battery monitoring mechanism, SOS button, buzzer, status LEDs, and MicroSD storage will provide the required sensing, emergency, alerting, and offline-storage functions. Each sensor and module will first be tested individually before being integrated into the complete onboard IoT device.

Following hardware integration, the embedded software will be developed to acquire sensor readings, process and filter sensor data locally, detect predefined emergency conditions, manage the manual SOS mechanism, activate local buzzer and LED alerts, and store relevant information locally when communication is unavailable. The integrated device will then be connected to the wider SeaGuardian system for communication and data synchronization.

Controlled laboratory simulations will be used to generate the experimental data required for evaluation. These simulations will include changes in boat tilt and motion, water leakage conditions, temperature variations, GPS location readings, battery conditions, manual SOS activation, and simulated network failures. Sensor readings and system responses generated during these experiments will be recorded for analysis.

An **iterative prototyping approach** will be followed during development. Hardware and software functions will be implemented and tested incrementally, allowing identified problems to be corrected before subsequent integration. Sensor calibration and repeated testing will be performed throughout the development process.

The final integrated prototype will be evaluated based on sensor functionality and accuracy, emergency response behaviour, power consumption, offline data handling, and overall system reliability. The experimental results will be used to determine whether the proposed SeaGuardian IoT Device can provide reliable onboard monitoring and edge-processing functionality under controlled simulated fishing-boat conditions.

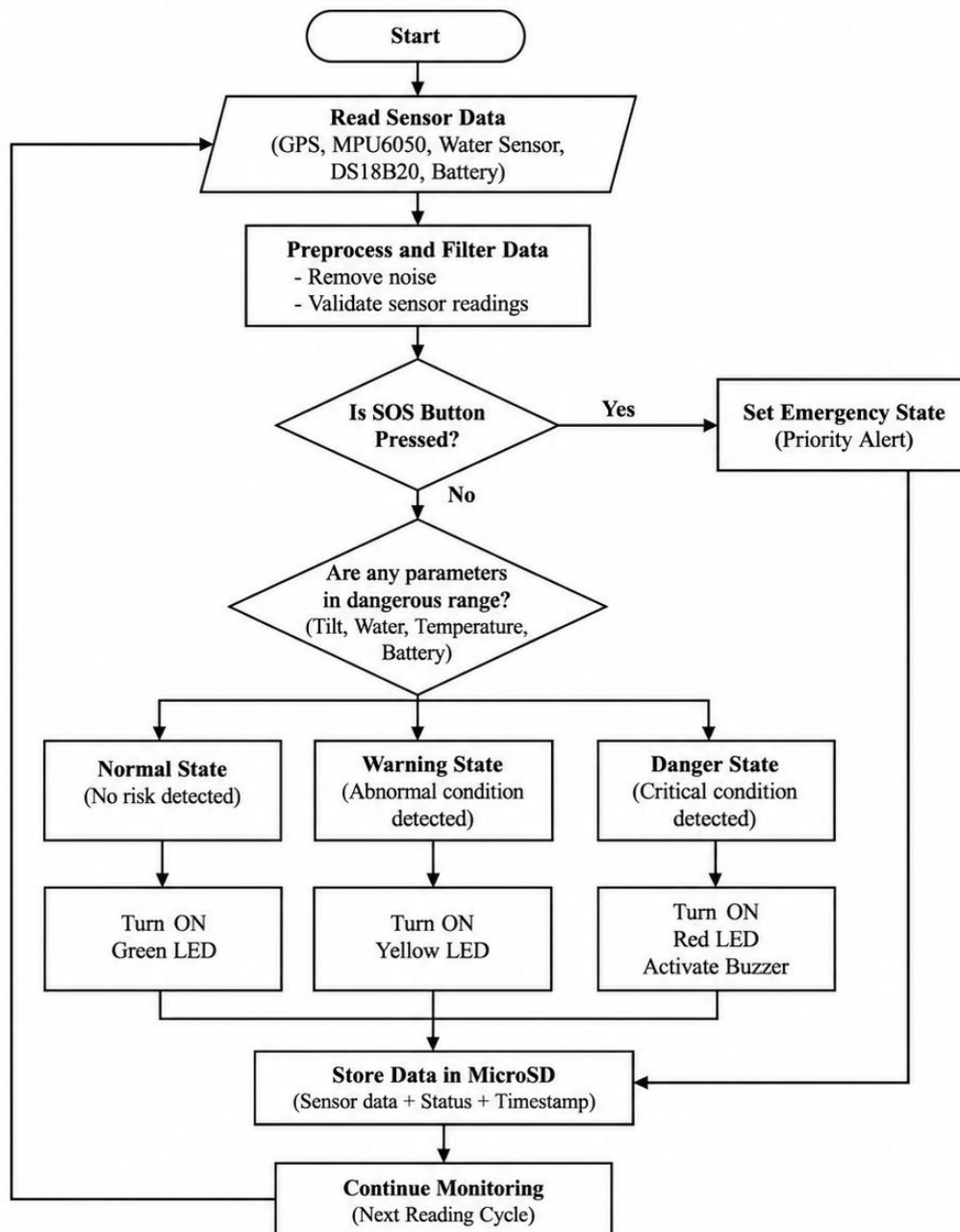


Figure 1-Edge Processing and Emergency Detection Flowchart of the SeaGuardian IoT Device

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3.2 Data / Participants / Experimental Inputs

This research will primarily use **experimental sensor data generated by the SeaGuardian IoT Device through controlled laboratory simulations**. Human participants are not required for the technical evaluation of the proposed component. The experimental inputs will represent normal and potentially hazardous onboard conditions that the device is intended to monitor.

The **NEO-6M GPS module** will provide geographical location information for testing the device's location-tracking functionality. GPS readings collected during controlled testing will be used to verify whether location information can be successfully acquired and processed by the ESP32.

The **MPU6050 sensor** will generate motion and orientation-related readings for monitoring boat tilt and movement. Different physical orientations and movements of the prototype will be introduced during laboratory testing to simulate changes in boat position and motion. The collected readings will be used to evaluate the device's ability to acquire and process motion-related information.

The **water leakage sensor** will generate readings under simulated dry and water-contact conditions. These experimental inputs will be used to determine whether the embedded device can distinguish between the tested conditions and trigger the required processing and alert behaviour.

The **DS18B20 temperature sensor** will provide temperature measurements under controlled temperature variations. The collected readings will be used to evaluate temperature-monitoring functionality and verify that temperature information can be correctly acquired, processed, stored, and transferred by the IoT device.

The **battery monitoring mechanism** will provide information related to the device's power condition. Different battery conditions will be observed during testing to evaluate whether the system can monitor and report its power status.

The **SOS push button** will provide a manual emergency input. Multiple controlled activations will be performed to verify that an SOS event is recognized by the embedded system and given the required priority for local alert and subsequent communication processes.

In addition to sensor inputs, **simulated network failures** will be introduced to evaluate offline operation. During communication unavailability, relevant sensor and emergency data will be stored locally using the MicroSD storage mechanism. The stored information can subsequently be used to verify offline data retention and synchronization behaviour after communication is restored.

Experimental data will be collected through repeated controlled tests and recorded for subsequent analysis. Before integrated experiments are conducted, each sensor and hardware module will be individually tested and calibrated where applicable. The resulting experimental dataset will contain sensor readings, emergency events, and system-response information generated under the defined laboratory conditions.

3.3 Baseline / Benchmark

The proposed SeaGuardian IoT Device will be evaluated against a **basic monitoring and emergency communication approach without integrated multi-sensor edge processing and offline data handling**. This baseline reflects the limitations identified in the existing fisherman safety approaches reviewed during the initial research, which primarily focus on GPS-based location tracking and manual emergency communication.

The baseline comparison will focus on the functional capabilities required for onboard safety monitoring. A conventional approach depends largely on the fisherman manually recognizing a hazardous situation and communicating the emergency through a mobile phone or radio system. Similarly, the existing IoT-based systems identified during the initial investigation mainly provide GPS tracking and emergency alert functionality, with limited integration of multiple onboard safety sensors.

In contrast, the proposed SeaGuardian IoT Device will integrate GPS location tracking, boat tilt and motion monitoring, water leakage detection, temperature sensing, battery monitoring, and manual SOS activation within a single ESP32-based embedded platform. The device will additionally provide local edge processing, buzzer and LED alerts, and MicroSD-based offline data storage.

The comparison will therefore consider whether the proposed device provides additional onboard safety capabilities compared with the identified baseline, particularly in terms of **multi-sensor monitoring, local processing, emergency detection and alerting, and operation during communication unavailability**.

At the individual hardware level, sensor outputs will also be compared against known or controlled test conditions where appropriate. For example, water detection will be tested using controlled dry and water-contact conditions, SOS functionality will be evaluated using known manual activations, and motion monitoring will be evaluated using controlled changes in prototype orientation. These controlled conditions will provide reference inputs for determining whether the developed device responds correctly.

The selected baseline is appropriate because the objective of this research is not only to provide location tracking or manual emergency communication, but to investigate the feasibility of an integrated onboard IoT device capable of continuously collecting and locally processing multiple safety-related inputs while maintaining essential data-handling capabilities during network interruptions.

3.4 Evaluation Metrics

The SeaGuardian IoT Device will be evaluated using quantitative and functional metrics that assess the performance of the individual sensors, emergency mechanisms, edge-processing functions, offline data handling, power consumption, and overall integrated prototype. The selected metrics are directly related to the objectives of the proposed component and will be measured through controlled laboratory experiments.

Table 1-Proposed Evaluation Metrics

Evaluation Metric	Why It Is Relevant	Target / Comparison
Sensor Functionality	Determines whether each integrated sensor correctly acquires and provides data to the ESP32.	Successful operation under defined controlled test conditions.
Sensor Accuracy	Determines how closely sensor readings represent known or controlled test conditions.	Compare measured readings with reference or controlled values where applicable.
Emergency Detection Response	Determines whether the device correctly responds to simulated hazardous conditions.	Correct detection and activation of the appropriate local alert during controlled emergency tests.
SOS Response	Evaluates whether manual emergency activation is correctly detected and prioritized.	Successful recognition of controlled SOS activations and activation of the required alert mechanism.
Offline Data Storage	Determines whether important sensor and emergency information can be retained when communication is unavailable.	Verify that relevant data is successfully stored on MicroSD during simulated network failure.
Data Recovery / Synchronization	Evaluates whether locally retained information remains available for subsequent synchronization after connectivity is restored.	Compare stored records with the data available after communication recovery.
Power Consumption	Determines the electrical power requirements of the integrated IoT device.	Measure and compare power usage during relevant device operating conditions.
System Reliability	Determines whether the integrated prototype operates consistently during repeated controlled experiments.	Compare successful operations with the total number of performed test trials.

Sensor accuracy will be evaluated where suitable reference or controlled conditions can be established. The MPU6050 will be evaluated using controlled changes in prototype orientation and movement, while the water leakage sensor will be evaluated using defined dry and water-contact conditions. Temperature measurements obtained from the DS18B20 will be evaluated against controlled or reference temperature measurements where an appropriate reference is available. GPS functionality will be evaluated by examining successful acquisition and processing of location information.

Emergency-response functionality will be evaluated by introducing controlled sensor conditions and manually activating the SOS mechanism. The resulting system behaviour, including local processing and buzzer and LED alert activation, will be recorded and compared with the expected response for each test condition.

Offline operation will be evaluated by intentionally simulating communication unavailability while the IoT device continues to acquire sensor information. The ability of the device to retain relevant information using MicroSD storage will then be verified. Where integration permits, the stored records will subsequently be checked after communication is restored to evaluate data recovery and synchronization behaviour.

Power consumption will be measured during relevant operating conditions to understand the energy requirements of the integrated prototype. Finally, overall system reliability will be assessed through repeated controlled tests of the integrated hardware and embedded software functions.

3.5 Validation Strategy

The proposed SeaGuardian IoT Device will be validated through **controlled laboratory testing, individual sensor verification, repeated experimental trials, and integrated prototype testing**. The validation process will be performed progressively, beginning with individual hardware components and continuing until the complete embedded device is integrated with the wider SeaGuardian system.

Initially, each sensor and hardware module will be tested independently before system integration. The NEO-6M GPS module will be tested for successful location-data acquisition, while the MPU6050 will be tested under controlled changes in orientation and movement. The water leakage sensor will be tested under defined dry and water-contact conditions, and the DS18B20 will be tested using controlled temperature conditions. Battery monitoring, MicroSD storage, buzzer, LED indicators, and the SOS push button will also be functionally verified before full integration.

Where appropriate, sensor readings will be compared with known or controlled reference conditions. Repeated measurements will be collected to identify inconsistent readings and evaluate the stability of the sensing functions. Sensor calibration and filtering will be performed where required before the final integrated experiments.

Emergency functionality will be validated using simulated emergency scenarios. These experiments will include controlled SOS activations and simulated hazardous sensor conditions. The expected system response will be compared with the actual response of the prototype, including sensor processing, emergency-state identification, and activation of local buzzer and LED alerts.

Offline operation will be validated by simulating communication failures while the device continues collecting sensor information. The MicroSD storage mechanism will be inspected to verify that relevant data is retained during the communication interruption. After communication is restored, the stored information will be checked for availability for subsequent synchronization with the wider SeaGuardian system.

Following individual component validation, the complete IoT prototype will undergo repeated integrated laboratory tests. The results will be analyzed using the defined evaluation metrics, including sensor functionality and accuracy, emergency response behaviour, offline data handling, power consumption, and overall system reliability.

The final validation will determine whether the developed SeaGuardian IoT Device satisfies the defined research objectives and can reliably perform its intended role as the **onboard sensing, emergency-input, local-processing, and offline data-storage component** of the complete SeaGuardian solution.

3.6 Development Approach

An **iterative prototyping development approach** will be used for the development of the SeaGuardian IoT Device. This approach is suitable because the proposed component involves the integration of multiple hardware sensors, embedded software, local data processing, emergency mechanisms, offline storage, and communication with other components of the complete SeaGuardian system. Developing and testing these functions incrementally allows hardware and software problems to be identified before the complete prototype is integrated.

The first iteration will focus on establishing the ESP32-based core embedded platform and verifying the basic operation of the selected hardware components. Each sensor and module will initially be connected and tested independently to confirm that the ESP32 can correctly acquire and process the required data.

The next iteration will focus on integrating the **NEO-6M GPS module, MPU6050, water leakage sensor, DS18B20 temperature sensor, and battery monitoring mechanism**. Sensor acquisition functions will be developed and tested individually before combining the readings within the main embedded system.

The following iteration will implement the **emergency and local alert mechanisms**. The SOS push button will be integrated as a manual emergency input, while the buzzer and LED indicators will provide local status and emergency alerts. The embedded software will then be extended to process sensor information locally and respond to predefined operating and emergency conditions.

A subsequent iteration will introduce **MicroSD-based offline data storage and edge data processing**. Sensor readings and relevant emergency events will be processed locally, with required information retained when communication is unavailable. Sensor filtering and data-processing functions will also be refined during this stage.

After the main hardware and software functions have been integrated, the prototype will undergo **calibration, debugging, and controlled laboratory testing**. Identified hardware, sensor, or software issues will be corrected through further iterations until the required functions operate consistently under the defined experimental conditions.

The final iteration will focus on integration with the wider SeaGuardian system and evaluation of the completed IoT device. The prototype will be tested for sensor functionality and accuracy, emergency response behaviour, offline data handling, power consumption, and overall system reliability. A suitable waterproof enclosure will also be considered for protecting the embedded hardware in the intended marine operating environment.

The iterative prototyping approach therefore enables the SeaGuardian IoT Device to be developed progressively through repeated cycles of **design, implementation, testing, evaluation, and refinement**, reducing integration risks while supporting systematic validation of the individual component.

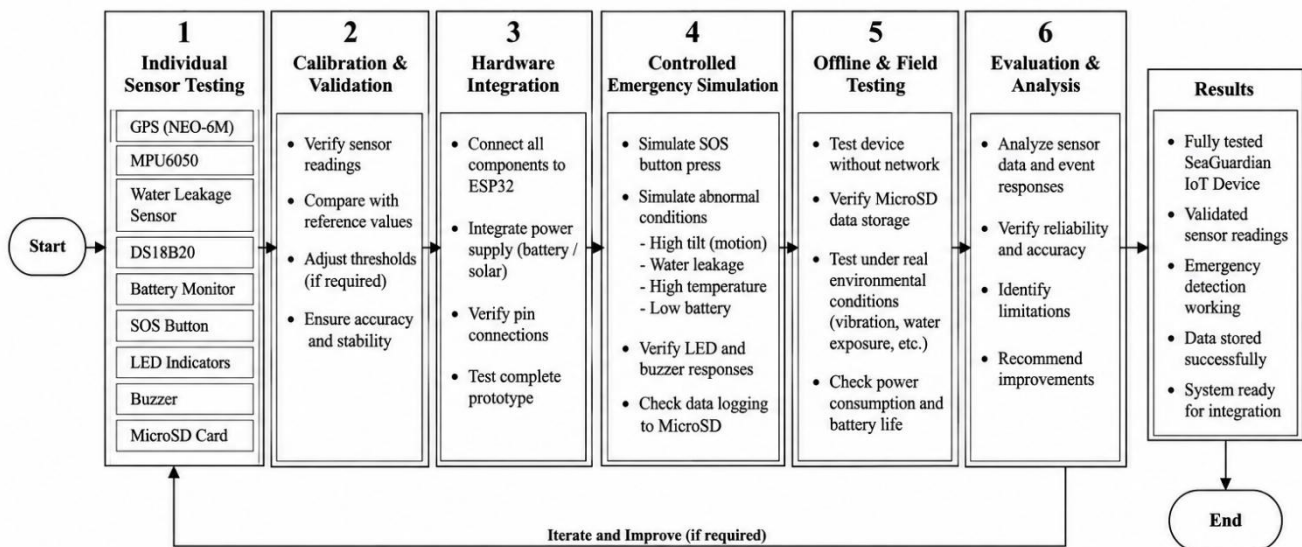


Figure 2-Experimental Methodology and Testing Process of the SeaGuardian IoT Device

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4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution

SeaGuardian is an AI-driven IoT-based smart fisherman safety and emergency communication system designed to provide real-time monitoring, intelligent risk assessment, and emergency response support for small-scale fishing activities. The complete solution integrates onboard IoT sensing, hybrid communication, cloud-based services, Digital Twin visualization, AI-based risk assessment, mobile rescue functionality, boat health monitoring, and fish storage monitoring.

The **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing** is the individual component addressed by this research. It operates as an onboard sensing and local-processing layer of the complete SeaGuardian architecture.

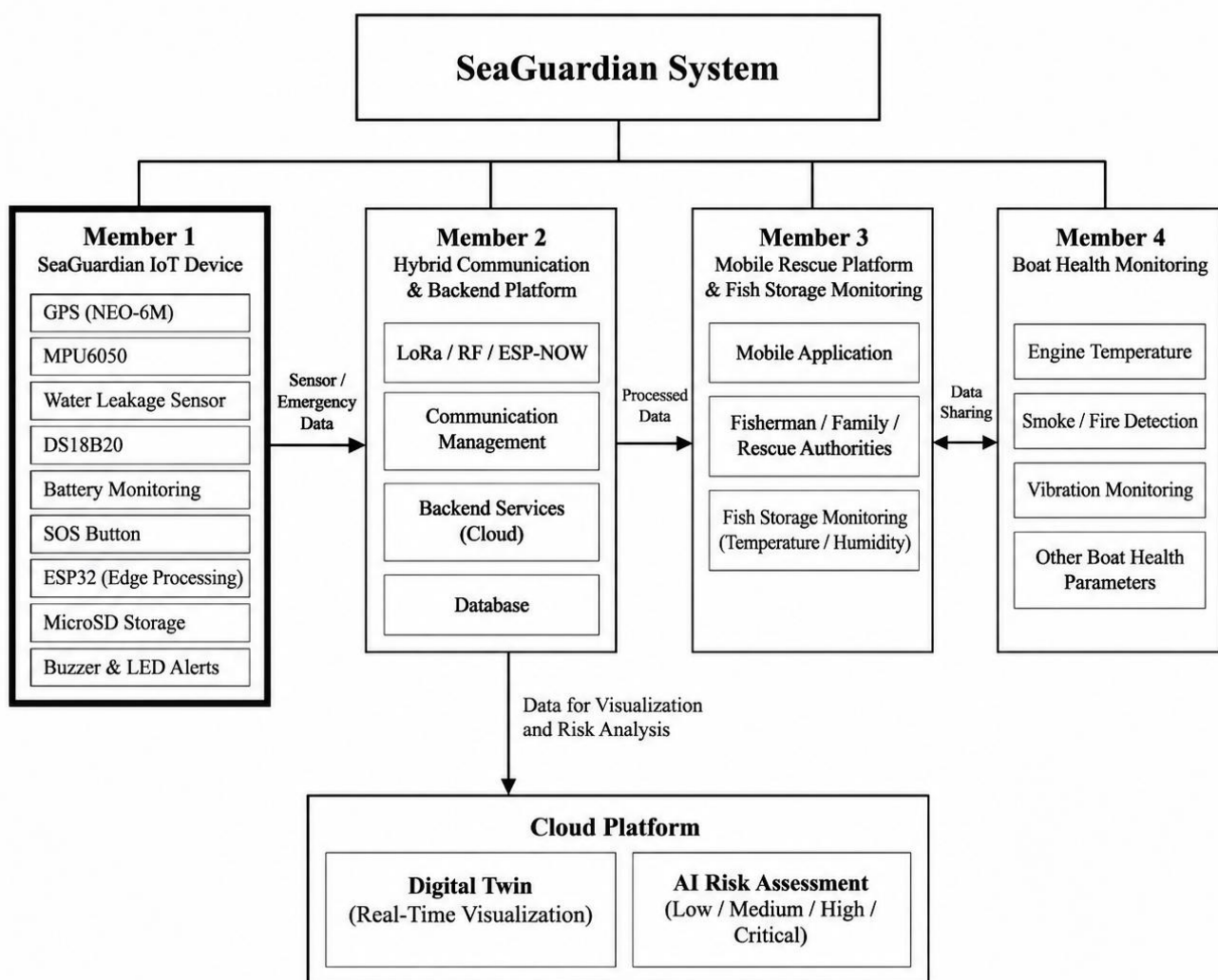


Figure 3-High-Level System Architecture of the SeaGuardian System

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The device is based on an **ESP32 microcontroller**, which acts as the central embedded processing unit. Multiple sensing and emergency modules are connected to the ESP32 to collect information related to the boat and its operating environment. The NEO-6M GPS module provides location information, while the MPU6050 provides motion and orientation-related information for monitoring boat tilt and movement. A water leakage sensor is used to identify water entering the monitored area, and a DS18B20 sensor provides temperature information. A battery monitoring mechanism is included to monitor the power condition of the device.

A manual **SOS push button** provides fishermen with a direct mechanism for initiating an emergency event. Local buzzer and LED indicators provide immediate onboard status and alert indications. These local mechanisms are important because an onboard emergency indication should not depend entirely on the availability of an external communication network.

The ESP32 performs local edge-processing functions on the acquired sensor information. Sensor readings can be processed and filtered before being transferred to other SeaGuardian components. This reduces the dependency on continuous remote processing and allows essential onboard monitoring functions to continue when external connectivity is limited.

The device also incorporates **MicroSD-based offline data storage**. When communication is unavailable, relevant sensor and emergency information can be retained locally. This information can subsequently be made available for synchronization when communication is restored.

The SeaGuardian IoT Device therefore provides the primary onboard data required by other parts of the complete SeaGuardian platform. Its outputs can be transferred through the project's communication infrastructure for further use by the cloud platform, Digital Twin, AI-based risk assessment, mobile rescue platform, and related monitoring functions.

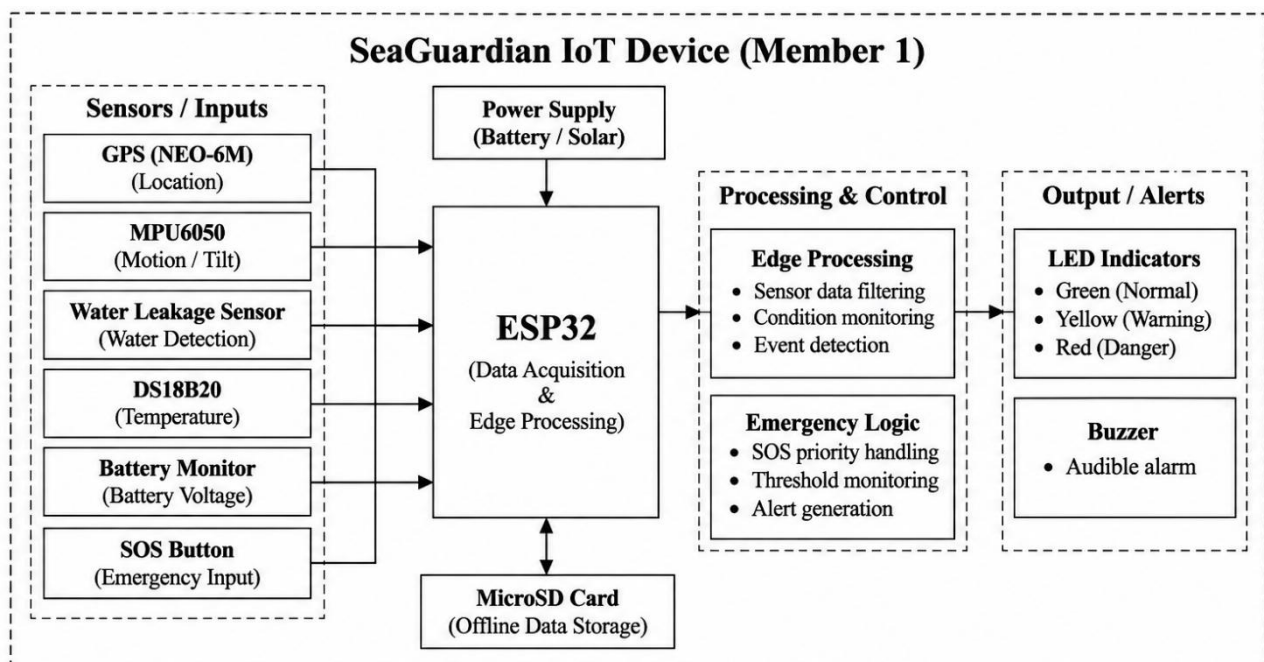


Figure 4-Architecture of the SeaGuardian IoT Device (Member 1)

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4.2 Overall Integration

The SeaGuardian solution is designed as a collection of interconnected components in which the onboard IoT device acts as an important source of real-time sensor and emergency information.

The SeaGuardian IoT Device collects and locally processes GPS, motion, water leakage, temperature, battery, and SOS information. After local processing, the required information is provided to the **Hybrid Emergency Communication System**, which is responsible for transferring information using the communication technologies defined within the complete SeaGuardian architecture.

The communicated information can subsequently be synchronized with the backend and cloud platform. The wider SeaGuardian system uses this information for functions including Digital Twin visualization and AI-based risk assessment. The mobile rescue platform can also use the resulting information to support monitoring and emergency-related functions for its intended users.

The integration therefore follows a general information flow of:

Onboard Sensors and Emergency Inputs → ESP32 Data Acquisition → Edge Processing → Local Alert / Offline Storage → Communication Infrastructure → Backend / Cloud Platform → Digital Twin, AI Risk Assessment and Mobile Monitoring

The primary dependency of this individual component is the communication interface required to transfer processed sensor information to the wider SeaGuardian platform. Conversely, the communication, visualization, and risk-analysis components depend on the SeaGuardian IoT Device to provide relevant onboard sensor and emergency information.

This separation of responsibilities allows the IoT device to continue performing its core sensing, local-processing, local-alerting, and offline-storage functions even when external communication is temporarily unavailable.

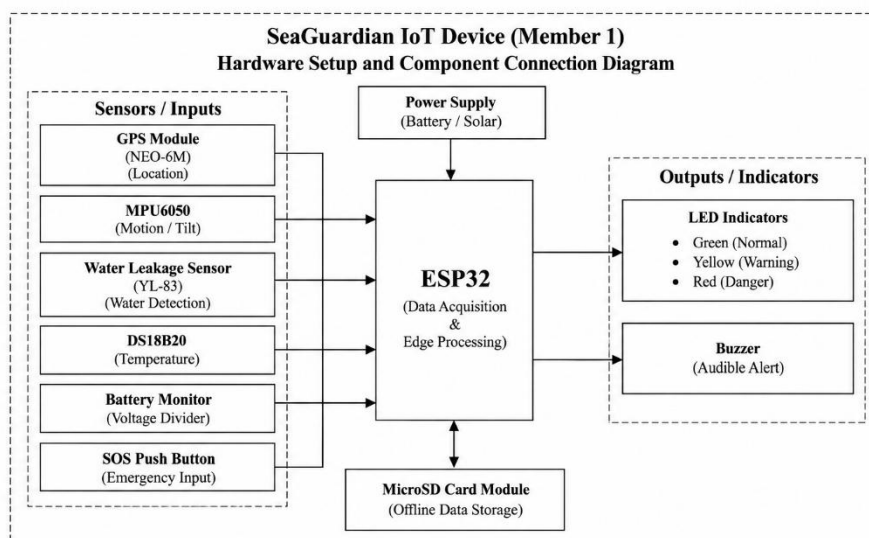


Figure 5-Hardware Setup and Component Connection Diagram

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4.3 Individual Responsibility Boundary

The scope of this individual research is limited to the design, development, integration, and evaluation of the **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing**. The development of the hybrid communication infrastructure, cloud platform, Digital Twin, AI risk assessment, mobile application, fish storage monitoring, and boat health monitoring are outside the primary responsibility of this individual component, except where integration interfaces are required.

Table 2-Individual Responsibility Boundary

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
ESP32-based SeaGuardian IoT Device	✓		
NEO-6M GPS integration	✓		
MPU6050 tilt and motion monitoring	✓		
Water leakage detection	✓		
DS18B20 temperature monitoring	✓		
Battery monitoring	✓		
SOS emergency button	✓		
Buzzer and LED alert mechanisms	✓		
MicroSD offline data storage	✓		
Edge data processing and sensor filtering	✓		
Hardware calibration and testing	✓		
Waterproof hardware enclosure	✓		
Power consumption and device reliability evaluation	✓		
Integration with communication infrastructure			Member 2
Hybrid LoRa, RF and ESP-NOW communication			Member 2
Cloud platform and backend			Member 2
Digital Twin and AI-based risk assessment			Member 2
Mobile rescue platform and fish storage monitoring			Member 3
Boat health monitoring and predictive fault detection			Member 4
Complete SeaGuardian system integration and testing		✓	All Members

5. USER REQUIREMENTS

The user requirements for the SeaGuardian IoT Device are derived initially from the research problem, limitations identified in existing fisherman safety approaches, and the functional scope defined for the proposed SeaGuardian system. The primary intended users of the overall system are small-scale fishermen, while information generated by the onboard device is also intended to support other SeaGuardian components used for monitoring and emergency response.

The initial requirement analysis indicates that the onboard device should continuously collect safety-related information, provide location information, identify relevant changes in onboard conditions, support manual emergency activation, provide immediate local alerts, and maintain essential data when communication is unavailable. Since the proposed device is intended for a marine environment, reliability, power efficiency, and physical protection of the embedded hardware are also important design considerations.

The proposed requirements will be further validated through appropriate stakeholder or domain-expert feedback before final system evaluation. Findings obtained through such requirement-elicitation activities will be documented and used to refine the functional and non-functional requirements where necessary.

5.1 Functional Requirements

The proposed functional requirements for the SeaGuardian IoT Device are as follows:

FR01 – Location Tracking:

The system shall acquire location information using the NEO-6M GPS module.

FR02 – Boat Motion Monitoring:

The system shall acquire motion and orientation information using the MPU6050 to support monitoring of boat tilt and movement.

FR03 – Water Leakage Detection:

The system shall monitor the water leakage sensor and process changes associated with the tested water-contact conditions.

FR04 – Temperature Monitoring:

The system shall acquire temperature readings using the DS18B20 temperature sensor.

FR05 – Battery Monitoring:

The system shall monitor the power condition of the SeaGuardian IoT Device using the implemented battery monitoring mechanism.

FR06 – Manual SOS Activation:

The system shall provide an SOS push button that enables manual activation of an emergency event.

FR07 – Local Alert Generation:

The system shall activate the appropriate buzzer and LED indications according to the implemented status and emergency conditions.

FR08 – Edge Data Processing:

The system shall process and filter relevant sensor information locally using the ESP32 before transferring the required information to other SeaGuardian components.

FR09 – Offline Data Storage:

The system shall store relevant sensor and emergency information using MicroSD storage when communication is unavailable.

FR10 – Data Availability After Connectivity Recovery:

The system shall retain locally stored information so that it is available for subsequent synchronization when communication is restored.

FR11 – Communication Interface:

The system shall provide the required processed sensor and emergency information to the SeaGuardian communication component for transmission to the wider platform.

FR12 – System Status Indication:

The system shall provide local visual and/or audible indications of relevant device and emergency states.

5.2 Non-Functional Requirements

The proposed non-functional requirements define the expected quality characteristics of the SeaGuardian IoT Device.

NFR01 – Reliability:

The device should operate consistently during repeated controlled laboratory tests and continue performing its core onboard monitoring functions under the defined experimental conditions.

NFR02 – Availability During Connectivity Loss:

Core sensor acquisition, local processing, local alerting, and offline data-storage functionality should remain operational when external communication is temporarily unavailable.

NFR03 – Power Efficiency:

The embedded hardware and software should be designed to minimize unnecessary power consumption, and device power consumption should be measured during experimental evaluation.

NFR04 – Data Integrity:

Sensor and emergency information stored locally should remain readable and available for subsequent processing or synchronization.

NFR05 – Maintainability:

The hardware and embedded software should be modular enough to allow individual sensors or modules to be tested, calibrated, replaced, or updated without redesigning the complete device.

NFR06 – Environmental Protection:

The final prototype should use a suitable protective enclosure to reduce exposure of the embedded electronic components to the intended marine operating environment.

NFR07 – Emergency Responsiveness:

The device should process manual SOS activation and implemented emergency conditions without unnecessary dependency on remote cloud processing.

NFR08 – Interoperability:

The IoT device should provide sensor and emergency information in a form that can be integrated with the communication and backend components of the wider SeaGuardian system.

6. COMMERCIALIZATION PLAN

The SeaGuardian IoT Device has commercialization potential as an onboard safety and monitoring solution for small-scale fishing boats. The proposed product combines multiple safety-monitoring functions within a single embedded platform and is designed to operate in environments where continuous internet connectivity cannot always be guaranteed.

6.1 Target Market and Customer Persona

The primary target market for the SeaGuardian IoT Device is the **small-scale fishing sector**, particularly fishing operations that require an affordable method of monitoring onboard safety conditions and supporting emergency response.

The primary customer persona is a small-scale fishing-boat owner or operator who requires a practical onboard safety device capable of monitoring important operating conditions while at sea. Potential organizational users of the wider SeaGuardian solution may also include fisheries-related organizations and authorities involved in maritime monitoring and emergency response.

The initial commercialization focus will be on the Sri Lankan fishing context addressed by this research. Expansion to other markets would require further investigation of local operating environments, communication infrastructure, regulatory requirements, and user needs.

6.2 Value Proposition

The main value proposition of the SeaGuardian IoT Device is the integration of **multiple onboard safety-monitoring and emergency functions within a single low-cost embedded platform**.

Unlike a solution that provides only location tracking or manual emergency communication, the proposed device combines GPS location tracking, boat tilt and motion monitoring, water leakage detection, temperature sensing, battery monitoring, manual SOS activation, local buzzer and LED alerts, edge processing, and offline data storage.

A further value proposition is its ability to maintain essential onboard functions when external communication is unavailable. Local sensor acquisition, edge processing, emergency indication, and MicroSD-based storage reduce the device's dependency on continuous internet connectivity.

6.3 Revenue Model

A potential commercialization model is a **hardware-based product model**, where the SeaGuardian onboard device is supplied as an integrated safety-monitoring unit.

Within the wider SeaGuardian ecosystem, additional service-based models could potentially be investigated for functions involving cloud monitoring, Digital Twin visualization, AI-based risk assessment, and mobile monitoring. However, these services extend beyond the primary responsibility of this individual component and would require further market and financial validation before commercialization.

6.4 Cost Estimation

The cost of the individual SeaGuardian IoT Device will primarily consist of the ESP32 microcontroller, GPS module, motion sensor, water leakage sensor, temperature sensor, battery-monitoring components, SOS button, buzzer, LEDs, MicroSD module and storage, power-related components, wiring/connectors, prototyping materials, and a suitable protective enclosure.

The detailed prototype hardware cost and justification will be presented in **Section 7 – Budget and Justification**.

Cost reduction could be investigated during future product development through optimized component selection, integrated PCB development, simplified assembly, and larger-scale component purchasing.

6.5 Pricing Strategy

A final commercial selling price will not be established during the initial proposal stage because the current research focuses on prototype development and technical evaluation rather than full commercial deployment.

A future pricing strategy should consider the total hardware and manufacturing cost, enclosure and assembly costs, installation requirements, maintenance, software and cloud-service costs where applicable, and an appropriate operating margin. The affordability of the system for the intended small-scale fishing market should remain an important consideration when determining a final commercial price.

6.6 Competitive Advantage

The proposed SeaGuardian IoT Device differentiates itself from the existing approaches identified during the initial research through the integration of multiple safety functions into a single onboard embedded platform.

Its proposed competitive advantages include:

- integration of multiple onboard safety sensors;
- local edge processing using the ESP32;
- GPS-based location monitoring;
- manual SOS emergency activation;
- local buzzer and LED alerts;
- MicroSD-based offline data storage;
- continued core monitoring during communication interruptions; and
- integration with the wider SeaGuardian platform for communication, cloud services, Digital Twin visualization, AI-based risk assessment, and mobile monitoring.

The combination of onboard sensing, local processing, offline operation, and integration with a wider intelligent safety platform provides the primary proposed differentiation of the SeaGuardian IoT Device.

6.7 Intellectual Property Considerations

The SeaGuardian system will integrate commercially available hardware components and established communication and software technologies. Therefore, ownership and licensing conditions associated with third-party hardware, software libraries, frameworks, communication technologies, and cloud services must be considered during future commercialization.

Potential project-specific intellectual property may arise from the original integration architecture, embedded processing logic, emergency-detection mechanisms, data-management approach, physical product design, and integration of the SeaGuardian components. Any intellectual property claims should be evaluated against existing patents, publications, products, and institutional intellectual-property policies before commercial deployment.

Appropriate consideration should also be given to the protection of the **SeaGuardian** product name, software source code, technical documentation, and original product designs where applicable.

7. BUDGET AND JUSTIFICATION

The budget for the SeaGuardian IoT Device consists primarily of hardware and electronic components required to design, develop, integrate, and experimentally evaluate the prototype. Wherever possible, actual purchase costs have been used instead of estimated values. Additional components required during later stages of development will be added to the final project budget.

Table 3-Current Hardware Budget and Justification

Component / Item	Quantity	Cost (LKR)	Justification
ESP32-S Wi-Fi/Bluetooth Development Board	1	1,680	Main embedded controller responsible for sensor acquisition, local edge processing, emergency logic, storage control, and system integration.
NEO-6M GPS Module with Antenna	1	1,100	Provides geographical location information for real-time boat location monitoring.
MPU6050 Motion Sensor	1	730	Used to acquire motion and orientation data for monitoring boat tilt and movement.
DS18B20 Waterproof Digital Temperature Sensor	1	320	Provides temperature measurements for onboard monitoring and is suitable for prototype testing where moisture exposure may occur.
Water / Rain Drop Sensor Module	1	240	Used in the prototype to detect simulated water-contact or leakage conditions.
MB-102 Breadboard	2	500	Supports temporary circuit construction, sensor integration, testing, and modification during prototype development.
Male-to-Female Jumper Wire Pack	1	170	Used to connect sensors and modules to the ESP32 during prototype development.
Male-to-Male Jumper Wire Packs	2	440	Used for breadboard connections and hardware integration.
5 mm LED Pack	1	50	Provides visual indications for device status, warnings, and emergency conditions.
220 Ω Resistor Pack	1	50	Used primarily for current limiting and protection of LED circuits.
Push Switch	1	100	Used to implement the manual SOS emergency activation mechanism.
Buzzer	1	100	Provides an audible local warning during implemented emergency conditions.

Soldering Iron and Soldering Lead	1 set	1,440	Required for permanent electrical connections, assembly, and prototype hardware construction.
Current Total Expenditure		6,920 LKR	Verified expenditure for the prototype components purchased to date.

The **ESP32** represents the central processing platform of the proposed device and is therefore a core project expense. It provides the processing and interfacing capabilities required to integrate the sensing, emergency, alerting, storage, and communication-related functions of the prototype.

The **NEO-6M GPS, MPU6050, water sensor, and DS18B20** represent the primary sensing components currently purchased for the individual research component. Together, they support location acquisition, boat motion and orientation monitoring, simulated water leakage detection, and temperature monitoring.

The **SOS push switch, buzzer, LEDs, and resistors** support the emergency and local alerting functions of the prototype. These components allow the embedded device to receive manual emergency input and provide immediate onboard visual and audible indications.

The **breadboards and jumper wires** are required for incremental prototype development and individual sensor testing before permanent integration. The soldering equipment supports the transition from temporary prototype connections to more secure hardware assembly.

The current verified expenditure for the purchased components is **LKR 6,920**. This represents expenditure to date rather than the complete final project cost. Additional expenditure may be required for components identified in the approved design but not represented in the current purchase records, including the battery monitoring implementation, MicroSD storage, power supply or battery system, waterproof enclosure, connectors, PCB or permanent mounting materials, and other integration requirements.

No paid software licence cost has currently been identified for this individual component. Any future cloud or communication-related expenditure associated with integration into the complete SeaGuardian system should be recorded separately and assigned according to the relevant component responsibility.

Drive Link to all the Bills, [Click Here](#)

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) divides the development of the **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing** into manageable research, development, integration, testing, evaluation, and documentation activities. The structure ensures that each stage of the individual research component can be systematically completed and evaluated.

Table 4-Work Breakdown Structure for the SeaGuardian IoT Device

WBS ID	Main Activity	Sub-Activity / Task	Expected Deliverable
1.0	Research and Requirement Analysis	Review fisherman safety problems and existing IoT-based solutions	Research problem and background
1.1	Research and Requirement Analysis	Conduct literature review related to IoT fisherman safety, onboard monitoring and edge processing	Literature review
1.2	Research and Requirement Analysis	Identify limitations and research gap	Defined individual research gap
1.3	Research and Requirement Analysis	Define functional and non-functional requirements	Requirement specification
2.0	Hardware Design	Identify required hardware components and sensors	Hardware specification
2.1	Hardware Design	Select ESP32 as the main embedded controller	Processing platform
2.2	Hardware Design	Design sensor and module connections	Hardware integration design
2.3	Hardware Design	Plan power and battery-monitoring implementation	Power-monitoring design
3.0	Sensor Development	Integrate and test NEO-6M GPS module	Location-monitoring function
3.1	Sensor Development	Integrate and test MPU6050	Tilt and motion-monitoring function
3.2	Sensor Development	Integrate and test water leakage sensor	Water-detection function
3.3	Sensor Development	Integrate and test DS18B20	Temperature-monitoring function
3.4	Sensor Development	Implement battery monitoring	Battery-status monitoring
4.0	Emergency and Alert System	Implement SOS push-button functionality	Manual emergency mechanism
4.1	Emergency and Alert System	Implement LED status indicators	Visual alert mechanism

4.2	Emergency and Alert System	Implement buzzer alerts	Audible alert mechanism
4.3	Emergency and Alert System	Implement emergency-priority handling	Emergency-response logic
5.0	Edge Processing	Develop local sensor-data acquisition	Sensor acquisition module
5.1	Edge Processing	Implement sensor filtering and local processing	Edge-processing module
5.2	Edge Processing	Implement condition and emergency-processing logic	Local decision functionality
6.0	Offline Data Management	Integrate MicroSD storage	Local storage module
6.1	Offline Data Management	Store relevant sensor and emergency information	Offline data records
6.2	Offline Data Management	Test operation during simulated network failure	Offline-operation results
6.3	Offline Data Management	Verify availability of stored data after connectivity recovery	Data recovery/synchronization evidence
7.0	Prototype Integration	Integrate sensors, emergency controls, processing and storage	Integrated IoT prototype
7.1	Prototype Integration	Integrate with SeaGuardian communication component	Communication interface
7.2	Prototype Integration	Assemble hardware into suitable protective enclosure	Physical prototype
8.0	Testing and Calibration	Individually test and calibrate sensors	Sensor test results
8.1	Testing and Calibration	Conduct controlled normal-condition tests	Normal-operation results
8.2	Testing and Calibration	Conduct simulated emergency-condition tests	Emergency-test results
8.3	Testing and Calibration	Test simulated communication failure	Offline-operation results
9.0	Evaluation	Evaluate sensor functionality and accuracy	Sensor evaluation results
9.1	Evaluation	Evaluate emergency-response behaviour	Emergency-response results
9.2	Evaluation	Evaluate power consumption	Power-consumption results
9.3	Evaluation	Evaluate overall system reliability	Reliability results

10.0	Documentation	Record experimental results and evidence	Experimental documentation
10.1	Documentation	Analyze and discuss research findings	Results and discussion
10.2	Documentation	Prepare final research documentation	Final dissertation/report
10.3	Documentation	Prepare demonstration and presentation materials	Final demonstration and presentation

The WBS follows an incremental development process in which individual hardware components are first tested independently before integration into the complete SeaGuardian IoT Device. After hardware and embedded-software integration, controlled experiments will be conducted to evaluate the device according to the research objectives and evaluation metrics.

Integration with the communication component will be performed after individual sensor functionality has been verified. This reduces the possibility of communication-related issues interfering with the initial hardware validation process. The final stages will focus on integrated testing, evaluation, analysis, and documentation of the research findings.

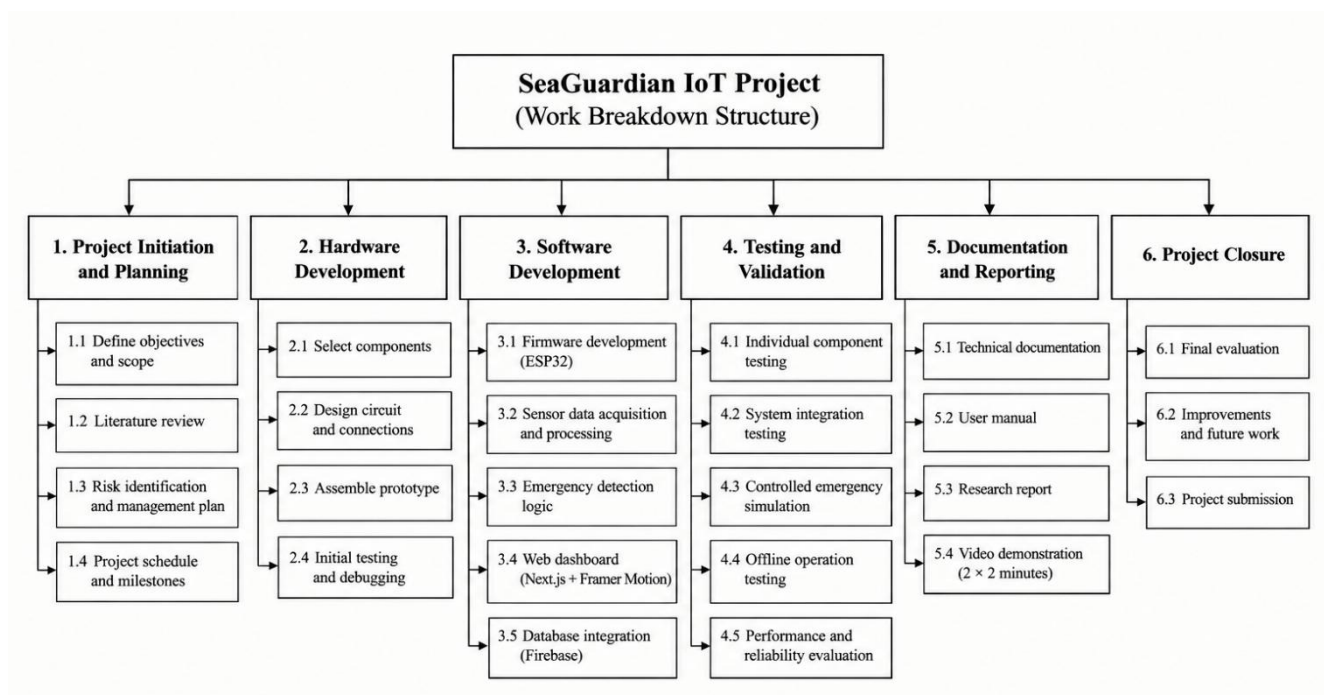


Figure 6-Work Breakdown Structure (WBS) of the SeaGuardian IoT Device (Member 1)

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9. GANTT CHART

The Gantt Chart presents the planned timeline for the research, development, integration, testing, evaluation, and documentation of the **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing**. The schedule extends from September 2026 to June 2027 and follows the activities identified in the Work Breakdown Structure.

Table 5-Gantt Chart for SeaGuardian IoT Device Development

Activity	Sep 26	Oct 26	Nov 26	Dec 26	Jan 27	Feb 27	Mar 27	Apr 27	May 27	Jun 27
Literature Review & Background Research	•	•	•							
Research Gap & Requirements Finalization	•	•								
Hardware Selection & System Design	•	•								
ESP32 Core Development	•	•	•							
GPS Module Integration & Testing		•	•							
MPU6050 Integration & Testing		•	•							
Water Sensor Integration & Testing		•	•							
DS18B20 Integration & Testing		•	•							
Battery Monitoring Implementation			•	•						
SOS, Buzzer & LED Alert Development		•	•	•						
Edge Processing & Sensor Filtering			•	•	•					
MicroSD Offline Storage				•	•					
Integrated IoT Prototype Development				•	•	•				
Protective Enclosure Development					•	•				
Integration with Communication Component						•	•			
Sensor Calibration & Controlled Testing						•	•			
Emergency Scenario Testing							•	•		
Network Failure / Offline Testing							•	•		
Power Consumption Evaluation								•		

Reliability Evaluation								•	•	
Data Analysis & Results Evaluation								•	•	
Dissertation / Research Documentation			•	•	•	•	•	•	•	•
Final Prototype Refinement									•	•
Final Demonstration & Presentation Preparation										•
Final Submission										•

Legend: • = Planned activity period

The development process begins with literature analysis, requirement identification, hardware selection, and initial ESP32 development. Individual sensors and emergency mechanisms will then be implemented and tested before being combined into the integrated SeaGuardian IoT Device.

Edge processing and offline storage will be developed after the primary sensing functions have been established. The integrated prototype will subsequently be connected to the SeaGuardian communication component and evaluated using controlled laboratory simulations.

The final stages of the research will focus on sensor calibration, simulated emergency scenarios, offline operation, power consumption, system reliability, analysis of experimental results, prototype refinement, dissertation preparation, and the final project demonstration.

10. REFERENCES LIST

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11. APPENDICES

The appendices contain supporting evidence and supplementary materials related to the design, development, and planned evaluation of the **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing**.

Appendix A – Proposal Evidence and Decision Log

The following Evidence and Decision Log records the major technical and research decisions made during the proposal stage and identifies the evidence used to support each decision.

Table 6- Proposal Evidence and Decision Log

Decision ID	Claim / Need	Evidence Source	Design / Research Decision	Proposal Section
ED01	The onboard device requires a central embedded processing platform.	Approved SeaGuardian TAF and prototype development requirements	ESP32 selected as the main controller for sensor acquisition and local edge processing.	Methodology / Architecture
ED02	Location information is required as part of the onboard safety-monitoring system.	Approved SeaGuardian TAF	NEO-6M GPS module selected for location acquisition.	Methodology / Architecture
ED03	Boat movement and orientation should be monitored.	Approved SeaGuardian TAF	MPU6050 selected for tilt and motion monitoring.	Methodology / Architecture
ED04	Water entering the monitored area should be detectable during prototype experiments.	Approved SeaGuardian TAF	Water/rain-drop sensor module selected for controlled water-contact testing.	Methodology
ED05	Temperature is one of the onboard parameters included within the individual component.	Approved SeaGuardian TAF	DS18B20 selected for temperature monitoring.	Methodology / Architecture
ED06	Fishermen require a mechanism for manually initiating an emergency event within the proposed system.	Approved SeaGuardian TAF	SOS push button included as a manual emergency input.	Requirements / Architecture
ED07	Immediate onboard indication is required for implemented warning and emergency states.	Approved SeaGuardian TAF	Buzzer and LED indicators included for local audible and visual alerts.	Requirements / Architecture

ED08	Sensor information should be processed locally before transmission.	Approved SeaGuardian TAF	Edge processing and sensor filtering will be implemented on the ESP32.	Research Gap / Methodology
ED09	Relevant data should be retained when communication is unavailable.	Approved SeaGuardian TAF	MicroSD-based offline storage included in the proposed architecture.	Research Gap / Methodology
ED10	The prototype requires systematic technical validation.	Approved SeaGuardian TAF	Controlled laboratory simulations, individual sensor testing, calibration, power evaluation, and reliability testing selected for evaluation.	Methodology
ED11	Prototype development requires physical electronic components and construction materials.	Actual purchase receipts	Hardware components and prototyping materials were purchased for implementation and experimental testing.	Budget

Appendix B – Prototype Hardware Purchase Evidence

This appendix provides supporting purchase evidence for the hardware components acquired during the initial prototype development of the SeaGuardian IoT Device. The expenditure presented here corresponds to the hardware costs recorded in Section 7 of this proposal.

The purchased components include the ESP32-S development board, NEO-6M GPS module, MPU6050 sensor, DS18B20 temperature sensor, water/rain sensor, MB-102 breadboards, jumper wires, LEDs, resistors, push switch, buzzer, and soldering equipment required for initial hardware development and integration.

Based on the available purchase evidence, the current verified expenditure for the initial prototype hardware is **LKR 6,920**. This amount represents expenditure incurred to date and should not be interpreted as the final total cost of the complete SeaGuardian IoT Device.

Additional components required during subsequent development stages may include a battery-monitoring module, MicroSD storage module, battery and power-management components, waterproof enclosure, connectors, PCB or permanent mounting materials, and other integration components. These costs will be recorded separately when the relevant components are selected and purchased.

The following purchase receipts and supporting evidence correspond to the hardware expenditure reported in Section 7.

Drive Link to all the Bills, [Click Here](#)

Appendix C – System Architecture

This appendix will contain the final black-and-white technical architecture diagram of the SeaGuardian system, highlighting the **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing** and its integration boundary with the other SeaGuardian components.

Appendix D – Prototype Development And Hardware Integration Evidence

This appendix presents available evidence from the initial prototype development and hardware integration activities conducted for the SeaGuardian IoT Device. The prototype is based on the ESP32 microcontroller, which functions as the main onboard processing unit for sensor data acquisition, local processing, emergency handling, and control of local alert mechanisms.

Initial prototype development focuses on the integration and individual testing of the hardware components required for the proposed onboard safety-monitoring system. These components include the NEO-6M GPS module for location information, MPU6050 for motion and orientation sensing, water leakage sensor for water detection, DS18B20 for temperature monitoring, SOS push button for manual emergency activation, LEDs for device-status indication, and a buzzer for local emergency alerts.

The hardware is initially assembled and tested using breadboard-based connections to allow individual components to be verified before permanent integration. Individual module testing is followed by progressive hardware integration with the ESP32. This development approach allows wiring, sensor readings, emergency controls, and output devices to be checked independently before evaluating the complete integrated prototype.

Prototype development evidence available at the proposal stage is presented below. Only hardware that has been assembled or tested is included as completed prototype evidence. Components and functions that have not yet been integrated are treated as planned development activities and are not presented as completed work.

Drive Link to Prototype Evidence, [Click Here](#)

Appendix E – Experimental Test Plan

The experimental test plan will document the controlled conditions used to evaluate the SeaGuardian IoT Device. Planned tests include GPS acquisition, controlled MPU6050 orientation and movement testing, water sensor dry/water-contact testing, DS18B20 temperature testing, battery monitoring, SOS activation, buzzer and LED response, MicroSD offline storage, simulated communication failure, power-consumption measurement, and integrated system reliability testing.

Test conditions, expected behaviour, actual results, and observations will be recorded during implementation and evaluation.

Appendix F – Work Breakdown Structure

The detailed Work Breakdown Structure for the individual research component will be included in this appendix where additional supporting detail is required.

Appendix G – Gantt Chart

The detailed project schedule extending to June 2027 will be included in this appendix where additional supporting detail is required.

Appendix H – Literature Comparison

The following table compares selected existing studies related to fisherman safety, vessel tracking, maritime-border detection, IoT-based monitoring, and intelligent maritime safety systems. The comparison focuses on the main approaches, technologies, contributions, limitations, and relevance of the reviewed studies to the proposed SeaGuardian IoT Device. This comparison supports the identification of the research gap addressed by the proposed individual research component.

Table 7-Literature Comparison of Existing Fisherman Safety Approaches

Reference	Existing Approach	Main Technologies	Main Contribution	Limitation Relevant to This Research	Relevance to SeaGuardian
[1] Amuthakkanna et al.	Review of fisherman-boat automation and intelligent technologies	AI, IoT and intelligent maritime technologies	Reviews technologies supporting fisherman safety, security, navigation, and information sharing	Provides a broad review of fisherman-boat automation rather than implementation of the integrated onboard device proposed in this research	Supports the need for integrated intelligent technologies to improve fisherman safety
[2] Fisherman Border Detector Using IoT and LoRa	Maritime-border detection and alert system for fishermen	IoT, LoRa and location-based technologies	Investigates the use of IoT and long-range communication for maritime-border detection and fisherman alerts	Primarily focuses on border detection and communication rather than comprehensive monitoring of multiple onboard safety conditions	Demonstrates the usefulness of location-aware alerts while motivating broader onboard safety monitoring
[3] Ramesh et al.	IoT-assisted fisherman aid for border detection and alerts	IoT and GPS	Uses GPS-based information to support fisherman location awareness and maritime-border alerts	Primarily emphasizes GPS-based location and border detection rather than integrated multi-sensor onboard	Supports the integration of GPS while motivating the addition of other onboard safety sensors

				condition monitoring	
[4] Gomathi et al.	SMS- and GPS-based fisherman tracking system	IoT, GPS and SMS	Demonstrates the use of location tracking and communication technologies to support fisherman safety	Mainly concentrates on fisherman tracking and communication rather than comprehensive onboard safety-condition monitoring	SeaGuardian extends beyond location tracking by integrating multiple onboard safety parameters and local processing
[5] Jothilingam and Glindis	IoT-based border-alert and secured system for fishermen	IoT, location-based monitoring and alert technologies	Provides maritime-border awareness and safety alerts for fishermen	Primarily addresses maritime-border safety rather than integrated environmental and vessel-condition monitoring	Supports the importance of emergency alerts while SeaGuardian investigates a broader multi-sensor onboard safety device

The literature comparison indicates that existing fisherman-safety research has substantially investigated GPS-based tracking, maritime-border detection, communication, and safety-alert mechanisms [1]–[5]. These studies demonstrate the value of IoT, GPS, wireless communication, and intelligent technologies for improving fisherman safety and maritime awareness.

However, the reviewed approaches provide comparatively limited integration of multiple onboard safety parameters and local processing functions within a single low-cost embedded device. In particular, the reviewed approaches do not address the complete combination of onboard location monitoring, boat motion and tilt monitoring, water leakage detection, temperature monitoring, battery-condition monitoring, manual emergency activation, local emergency alerts, edge processing, and offline data storage investigated in this research.

Therefore, the proposed **SeaGuardian IoT Device for Real-Time Onboard Monitoring and Edge Processing** investigates an ESP32-based integrated platform combining a NEO-6M GPS module, MPU6050 motion sensing, water leakage detection, DS18B20 temperature sensing, battery monitoring, SOS emergency activation, LED and buzzer alerts, local edge processing, and MicroSD-based offline data storage. This integrated approach is intended to provide continuous onboard safety monitoring while maintaining essential local monitoring and data-storage capabilities in fishing environments where network connectivity may be limited or intermittent.

Appendix I – AI Use Disclosure

Generative AI tools were used during the proposal preparation process to support activities such as language refinement, content organization, brainstorming, and assistance with structuring research documentation. AI-generated outputs were reviewed and adapted by the researcher before inclusion in the proposal.

Generative AI tools were not used as a replacement for experimental evidence, stakeholder evidence, research results, or academic sources. Technical claims, literature citations, experimental results, and evidence included in the final research work will be verified using appropriate original sources and actual project data.

Technical diagrams submitted as part of the proposal will be created manually in accordance with the proposal requirements and will not be generated using AI tools.

Activity	AI Use	Verification
Language refinement	Grammar and academic wording	Reviewed and edited by researcher
Content organization	Organizing sections and research content	Checked against proposal template and research scope
Brainstorming	Supporting research ideas and documentation structure	Relevant suggestions reviewed before use
Literature support	Assisting with literature organization and reference formatting	Sources checked against original publications
Technical diagrams & results	AI not used to create final diagrams, experimental results, or evidence	Based on manually created diagrams and actual project evidence